

Interband Absorption and PL in Lens-Shaped Quantum Dots: An Adiabatic Approach

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ABSTRACT

We present an analytic–numerical framework for carriers and excitons in lens-shaped (semi-ellipsoidal) quantum dots within the effective-mass and envelope-function approximations, assuming hard-wall confinement and neglecting strain, dielectric mismatch, and band-mixing. Exploiting the geometry ($c \ll a$), we use an adiabatic separation of fast (longitudinal) and slow (in-plane) motion to obtain closed-form single-particle states and energies; the eigenfunctions are purely geometric (independent of effective mass), while the energies scale inversely with the effective mass. The exciton transition energy is built from the band gap, the electron and hole confinement contributions, and the electron–hole Coulomb attraction. The interaction energy was evaluated numerically in first-order perturbation theory using the analytic envelopes. Interband absorption follows from bright-state selection rules, and photoluminescence is obtained from absorption via the van Roosbroeck–Shockley relation with Lorentzian broadening. For representative GaAs-like parameters, single-particle energies are much more sensitive to the dot height than to the lateral semi-axis, states with higher longitudinal quantum number lie well above the lowest branch, the Coulomb binding decreases with increasing size, showing comparable fractional sensitivity to both the vertical and lateral semi-axes and groups primarily by radial quantum number. The framework yields compact formulas, transparent scaling trends, and interpretable spectra and lifetimes for the design and analysis of lens-shaped quantum dots.

1. INTRODUCTION

Semiconductor quantum dots (QDs) are often described as “artificial atoms” because quantum confinement yields discrete energy spectra and size–tunable optical properties with broad impact across imaging, sensing, and optoelectronics. Recent surveys and application–driven studies highlight both the technological progress and remaining challenges in materials, devices, and integration.^{1,2} In bioimaging and biondiagnostics, QDs offer bright, photostable labels with multiplexing capabilities,³ while colloidal carbon QDs provide complementary processability and low–toxicity routes for photonic uses.⁴ Together, these works underscore the central role of QDs in modern photonics and the importance of tractable theoretical models that connect geometry to spectra.

Beyond composition and overall size, *shape* is a powerful design parameter for tailoring selection rules, oscillator strengths, and polarization characteristics. Ellipsoidal families—including oblate and prolate spheroids—have been analyzed in detail for their electronic states and interband absorption,^{5,6} as well as for related ellipsoidal/lens–like nanostructures with strong shape anisotropy.^{7–9} In epitaxial systems, *lens-shaped* (semi-ellipsoidal) QDs arise naturally and have been used to probe fundamental many–body and magneto–optical effects. In particular, a series of theoretical and experimental studies has examined Kohn’s theorem in ellipsoidal geometries and its (non)implementation under magnetic field, including demonstrations in Ge/Si dots with hole gas.^{10–12} Related work on asymmetric biconvex lens–shaped dots mapped the influence of external fields on interband and intraband optical properties.¹³

Against this backdrop, the present paper develops a minimal effective–mass, envelope–function description for carriers and excitons in lens–shaped (semi–ellipsoidal) QDs with hard–wall confinement. We derive closed–form single–particle states in an adiabatic (fast/slow) separation, compute first–order Coulomb corrections to exciton energies using analytic envelopes, and obtain interband absorption together with photoluminescence (PL) via the van Roosbroeck–Shockley detailed–balance relation.¹⁴ For concreteness and to facilitate comparison with

the III–V literature, we reference lifetime/oscillator–strength conventions common in excitonic modeling of nitride and arsenide nanostructures.¹⁵ Our goal is to provide compact formulas, transparent scaling trends, and interpretable spectra that clarify how lens geometry organizes bright transitions and Coulomb–induced splittings.

2. THEORY

2.1 Single-Particle Energies and States

The geometry of the lens-shaped quantum dot is modeled by a cylindrically symmetric semi-ellipsoid illustrated in Figure 1.

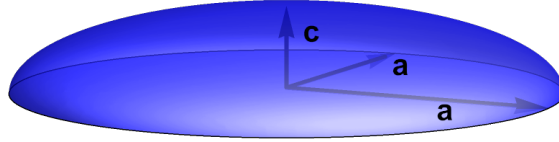


Figure 1. Schematic representation of a semi-ellipsoid with semi-axes a , a and c

The confining potential of a particle in such a quantum dot is defined as:

$$V(x, y, z) = \begin{cases} 0 & \frac{x^2+y^2}{a^2} + \frac{z^2}{c^2} < 1, \ z > 0 \\ \infty & \text{otherwise} \end{cases}. \quad (1)$$

where a and c denote the major and minor semi-axes of the semi-ellipsoid, respectively, subject to the condition $c \ll a$.

The single-particle Hamiltonian in cylindrical coordinates then takes the form

$$\hat{H} = -\left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \varphi^2} + \frac{\partial^2}{\partial z^2} \right) + V. \quad (2)$$

where distances are expressed in units of effective Bohr radius $r_B = \frac{\epsilon \hbar^2}{m^* e^2}$, where ϵ is the dielectric permittivity, and energy in units of the effective Rydberg energy $E_R = \frac{\hbar^2}{2m^* r_B^2}$.

From the geometry of the quantum dot and the condition $c \ll a$, it follows that the motion of the particle along the OZ axis is much faster than in the OXY plane, which makes it possible to apply the adiabatic approximation. Within this approximation, the particle Hamiltonian can be separated into two components:

$$\begin{aligned} \hat{H} &= \hat{H}_r + \hat{H}_z + V, \\ \hat{H}_r &= -\left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \varphi^2} \right), \\ \hat{H}_z &= -\frac{\partial^2}{\partial z^2}, \end{aligned} \quad (3)$$

where $\hat{H}_r$ describes the slow degrees of freedom, while $\hat{H}_z$ describes the fast ones.

The solution of the stationary Schrödinger equation is sought in the form

$$\psi(r, \varphi, z) = \psi_r(r) \psi_z(r, z) e^{im\varphi}. \quad (4)$$

For a fixed value of r , the fast subsystem can be represented as a potential well with an effective width

$$w(r) = c \sqrt{1 - \frac{r^2}{a^2}}. \quad (5)$$

The solution of the Schrödinger equation yields

$$\psi_z(r, z) = \sqrt{\frac{2}{w(r)}} \sin\left(\frac{\pi n_z z}{w(r)}\right), \quad \varepsilon_z(r) = \frac{\pi^2 n_z^2}{w^2(r)}, \quad (6)$$

where n_z is the quantum number of the fast subsystem. Here $\varepsilon_z(r)$ can be expanded in a Taylor series with respect to $\frac{r}{a}$ as

$$\varepsilon_z(r) \approx \alpha n_z^2 + \beta^2 n_z^2 r^2, \quad \alpha = \frac{\pi^2}{c^2}, \quad \beta = \frac{\pi}{ac}. \quad (7)$$

Substituting $\varepsilon_z(r)$ into the equation for the slow subsystem as an effective potential leads to the problem of a two-dimensional harmonic oscillator. Accordingly, $\psi_r(r)$, the radial component of the wave function, is given by

$$\psi_r(r) = \mathcal{A} r^{|m|} e^{-\beta n_z r^2/2} L_{n_r}^{|m|}(\beta n_z r^2), \quad (8)$$

where n_r and m are the radial and magnetic quantum numbers, respectively, $L_{n_r}^{|m|}(\cdot)$ is the associated Laguerre polynomial, and $\mathcal{A}$ is the normalization constant,

$$\mathcal{A} = \sqrt{\frac{(\beta n_z)^{|m|+1} n_r!}{\pi \Gamma(n_r + |m| + 1)}}. \quad (9)$$

By combining the radial and longitudinal components, we obtain the normalized wave function:

$$\psi_{n_z, n_r, m}(r, \varphi, z) = \sqrt{\frac{2(\beta n_z)^{|m|+1} n_r!}{\pi \Gamma(n_r + |m| + 1) w(r)}} r^{|m|} e^{-\beta n_z r^2/2} L_{n_r}^{|m|}(\beta n_z r^2) \sin\left(\frac{\pi n_z z}{w(r)}\right) e^{im\varphi}, \quad (10)$$

The corresponding single-particle energy spectrum is

$$E_{n_z, n_r, m} = \alpha n_z^2 + \beta n_z (2n_r + |m| + 1), \quad (11)$$

where $\alpha = \pi^2/c^2$ and $\beta = \pi/(ac)$.

For the lowest longitudinal branch $n_z = 1$, the energy depends only on $N \equiv 2n_r + |m|$; all states with the same N are degenerate with multiplicity $N + 1 = 2n_r + |m| + 1$ (with $m = 0$ counted once and $\pm m$ counted separately).

2.2 Exciton Energy

The Hamiltonian of the exciton is

$$\hat{H} = \hat{H}_e + \hat{H}_h - \frac{e^2}{\varepsilon |\mathbf{r}_e - \mathbf{r}_h|}, \quad (12)$$

where $\hat{H}_e$ and $\hat{H}_h$ are the single-particle Hamiltonians of the electron and hole, respectively, and the last term describes their Coulomb attraction. In this section, all lengths and energies are expressed in the effective atomic units introduced earlier (effective Bohr radius r_B and effective Rydberg E_R), in which the Coulomb term reads

$$-\frac{1}{|\mathbf{r}_e - \mathbf{r}_h|}. \quad (13)$$

When comparing with optical spectra, the relevant interband transition energy includes the band gap E_g :

$$E_{ex} = E_g + E_{eh}, \quad E_{eh} = E_e + E_h + V_{eh}, \quad (14)$$

where E_e and E_h are the confinement (single-particle) energies of the electron and hole measured from the band edges, and

$$V_{eh} = - \int \psi_e^*(\mathbf{r}_e) \psi_h^*(\mathbf{r}_h) \frac{1}{|\mathbf{r}_e - \mathbf{r}_h|} \psi_e(\mathbf{r}_e) \psi_h(\mathbf{r}_h) d^3\mathbf{r}_e d^3\mathbf{r}_h, \quad (15)$$

with $\psi_{e,h}$ the normalized envelopes from the previous subsection.

Under a uniform rescaling of the dot dimensions $(a, c) \rightarrow (\lambda a, \lambda c)$ with $\lambda > 0$, the normalized envelopes transform as $\psi_\lambda(\mathbf{r}) = \lambda^{-3/2} \psi(\mathbf{r}/\lambda)$. Inserting this into the definition of V_{eh} shows that $V_{eh} \rightarrow V_{eh}/\lambda$, while the single-particle terms scale as $E_{e,h} \rightarrow E_{e,h}/\lambda^2$. Since E_g is a material parameter, the total optical transition energy obeys

$$E_{ex}(\lambda) = E_g + \frac{E_e + E_h}{\lambda^2} + \frac{V_{eh}}{\lambda}. \quad (16)$$

2.3 Exciton Lifetime

The radiative lifetime of the exciton provides a key measure of its optical activity, as it determines the probability of spontaneous photon emission. In this subsection, we use SI units. The expression for the radiative lifetime is¹⁵

$$\tau_{ex} = \frac{2\pi \varepsilon_0 m_e c^3 \hbar^2}{\sqrt{\varepsilon} e^2 E_{ex}^2 f}, \quad (17)$$

where m_e is the free-electron mass and f is the oscillator strength. Within the Kane model, it is

$$f = \frac{E_p}{E_{ex}} \left| \int \psi_{ex}(\mathbf{r}, \mathbf{r}) d^3 \mathbf{r} \right|^2, \quad (18)$$

where E_p is the Kane energy (material parameter) and $\psi_{ex}(\mathbf{r}, \mathbf{r})$ is the exciton wave function at coincident coordinates.

Taking into account that $\psi_{n_z, n_r, m} = \psi_{n_z, n_r, -m}^*$ and using the orthonormality of the modes, the overlap integral can be evaluated as

$$\int \psi_e(\mathbf{r}) \psi_h(\mathbf{r}) d^3 \mathbf{r} = \delta_{n_z^e, n_z^h} \delta_{n_r^e, n_r^h} \delta_{m^e, -m^h}. \quad (19)$$

Thus the oscillator strength reduces to

$$f = \frac{E_p}{E_{ex}} \delta_{n_z^e, n_z^h} \delta_{n_r^e, n_r^h} \delta_{m^e, -m^h}. \quad (20)$$

For bright transitions ($n_z^e = n_z^h$, $n_r^e = n_r^h$, $m^e = -m^h$) this gives $f = E_p/E_{ex}$, and substituting into the lifetime formula yields

$$\tau_{ex}^{(\text{bright})} = \frac{2\pi \varepsilon_0 m_e c^3 \hbar^2}{\sqrt{\varepsilon} e^2 E_p E_{ex}}, \quad (21)$$

i.e., the radiative lifetime scales inversely with the transition energy. States violating the selection rules are dark in the dipole approximation. In practice, the experimentally observed lifetime may be further reduced by non-radiative channels.

2.4 Interband Absorption and Photoluminescence

Interband absorption probes the allowed optical transitions in the quantum dot and is strongly shaped by excitonic effects. In the envelope-function picture, the absorption coefficient (up to a material-dependent prefactor) can be written as a sum over electron-hole envelope states,

$$\alpha(\hbar\omega) \propto \sum_{n_z^e, n_r^e, m^e} \sum_{n_z^h, n_r^h, m^h} \left| \int \psi_{ex}(\mathbf{r}, \mathbf{r}) d^3 \mathbf{r} \right|^2 \delta(\hbar\omega - E_{ex}), \quad (22)$$

where $\hbar\omega$ is the photon energy and E_{ex} is the exciton transition energy for the corresponding electron-hole pair. Using the overlap result (19) from the previous subsection (bright-state selection rules), the double sum collapses to a sum over bright transitions only,

$$\alpha(\hbar\omega) \propto \sum_{n_r, n_z, m} \delta(\hbar\omega - E_{ex}(n_r, n_z, m)), \quad (23)$$

with (n_r, n_z, m) obeying $m_e + m_h = 0$ and matched radial/longitudinal quantum numbers.

The spontaneous emission (photoluminescence) spectrum is related to the absorption spectrum by the van Roosbroeck–Shockley detailed-balance relation:¹⁴

$$R(\hbar\omega) = \frac{n^2(\hbar\omega)^2}{\pi^2 c^2 \hbar^3} \alpha(\hbar\omega) \frac{f_c(1-f_v)}{f_v - f_c}, \quad (24)$$

where n is the refractive index, c is the speed of light, and f_c and f_v are the Fermi–Dirac occupation probabilities of the conduction and valence band states that depend on the lattice temperature.

Substituting (23) into (24) yields

$$R(\hbar\omega) \propto (\hbar\omega)^2 \sum_{n_r, n_z, m} \delta(\hbar\omega - E_{ex}(n_r, n_z, m)) \frac{f_c(1-f_v)}{f_v - f_c}. \quad (25)$$

In thermal equilibrium when the quasi Fermi levels of the conduction and valence bands are equal $F_c = F_v$, the ratio $\frac{f_c(1-f_v)}{f_v - f_c}$ reduces to the Planck factor $[e^{\hbar\omega/k_B T} - 1]^{-1}$, recovering the usual black-body weighting of the absorption spectrum. Under weak excitation ($f_c \ll 1$, $f_v \approx 1$), the factor is close to unity and $R(\hbar\omega)$ follows $\alpha(\hbar\omega)$ up to the $(\hbar\omega)^2$ density-of-states prefactor.

In practice, delta-peaks are broadened by coupling to phonons. A convenient replacement is a Lorentzian with half-width at half-maximum γ ,

$$\delta(\hbar\omega - E_{ex}) \Rightarrow \frac{1}{\pi} \frac{\gamma}{(\hbar\omega - E_{ex})^2 + \gamma^2}, \quad (26)$$

with typical γ in the range 0.5–5 meV. The resulting broadened spectra are evaluated numerically in the next section.

3. RESULTS AND DISCUSSION

Figure 2 shows the probability density $|\psi_{n_z, n_r, m}(r, \varphi, z)|^2$ described by (10) for several representative states in the lens-shaped dot. For each state, we also schematically plot the longitudinal and radial marginal probability densities that are proportional to $|\psi_z|^2$ and $|\psi_r|^2$, respectively. The expected nodal structure is evident: along z the longitudinal factor $\sin(\pi n_z z/w)$ produces $n_z - 1$ internal nodes and n_z lobes, while in the radial direction the Laguerre part yields exactly n_r radial nodes. The azimuthal quantum number m controls the central behavior via the $r^{|m|}$ prefactor: for $m = 0$ the density is maximal at $r = 0$, whereas for $|m| \geq 1$ it vanishes at $r = 0$ and forms ring-like maxima whose radius grows with $|m|$. Overall, the examples show clear localization of the particle near the dot center for the lowest (n_r, m) and the emergence of additional nodes as n_z or n_r increase.

The size dependence of the corresponding single-particle energies is summarized in Figure 3. Varying the height c (left panel) produces large shifts, whereas varying the lateral size a at fixed c (right panel) has a much weaker effect. This sensitivity hierarchy reflects the spectrum (11). States with $n_z > 1$ lie far above the $n_z = 1$ manifold as a direct consequence of the relation $a \gg c$, and thus will be disregarded in what follows. The lowest level shows only a very weak dependence on a since the dominant α -term is controlled solely by c . The level spacings are uniform and are inversely proportional to either a or c .

Figure 4 presents the electron–hole Coulomb contribution $|V_{eh}|$ for several $n_z = 1$ states as a function of the dot dimensions. As either the height c (left) or the lateral semi-axis a (right) increases, the binding weakens monotonically. Notably, the sensitivity to relative changes in a and in c is comparable over the plotted ranges: a given fractional change in a or c produces a similar fractional change in $|V_{eh}|$. An interesting systematic ordering emerges: curves group primarily by the radial quantum number n_r (across different m) and with an increasing n_r the binding energy decreases.

To estimate the binding energy at arbitrary (a, c) , one can match the target aspect ratio a/c to the corresponding curve in Fig. 4, then recover the value for (a, c) using the uniform-size scaling law (16).

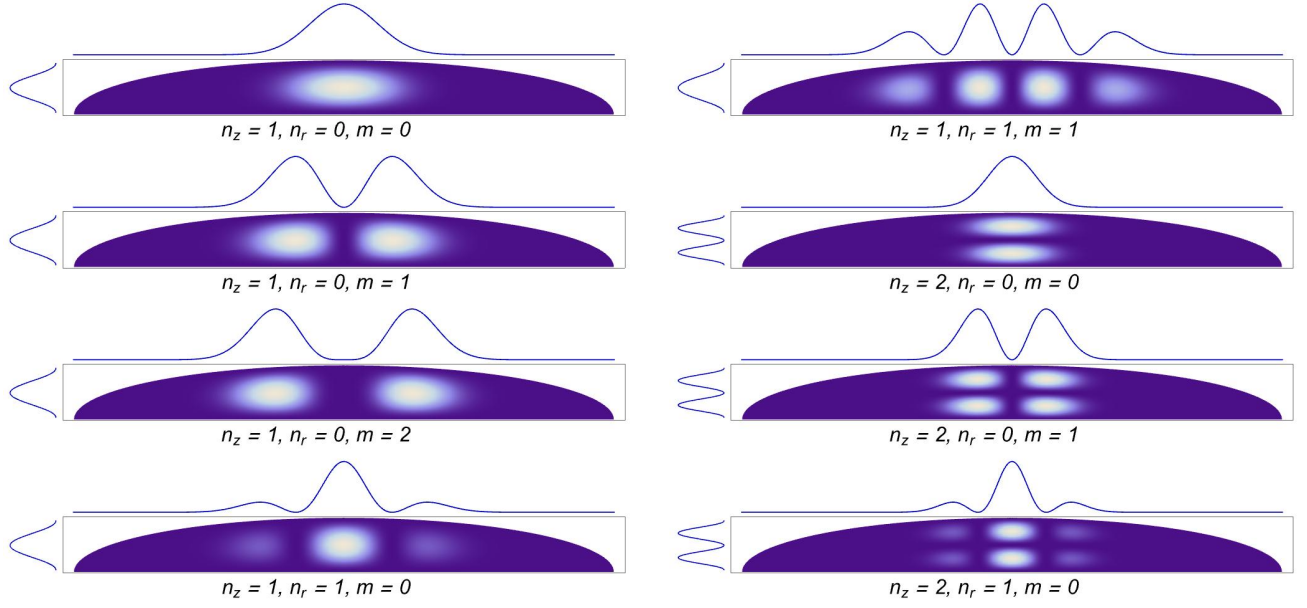


Figure 2. **Probability densities and marginals in the lens-shaped dot.** Each panel displays the probability density $|\psi_{n_z, n_r, m}(r, \varphi, z)|^2$ for a representative state. For each state, curves plotted to the left and on top show the longitudinal and the radial marginal probability densities respectively.

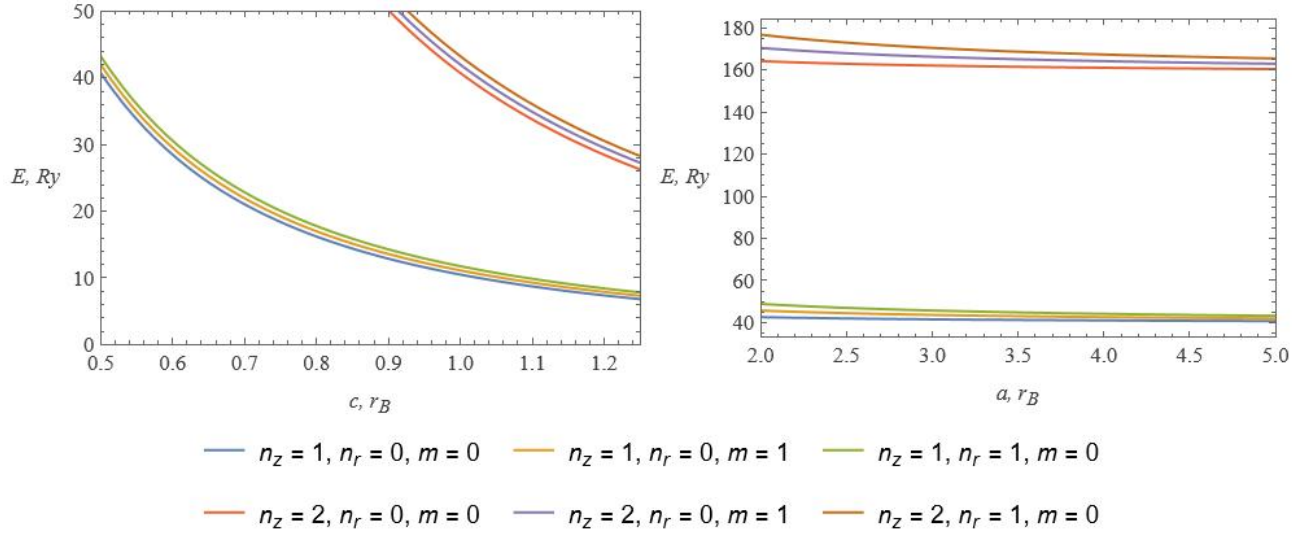


Figure 3. **Electron energies versus dot dimensions.** **Left:** dependence on the vertical semi-axis c at fixed $a = 5r_B$. **Right:** dependence on the lateral semi-axis a at fixed $c = 0.5r_B$.

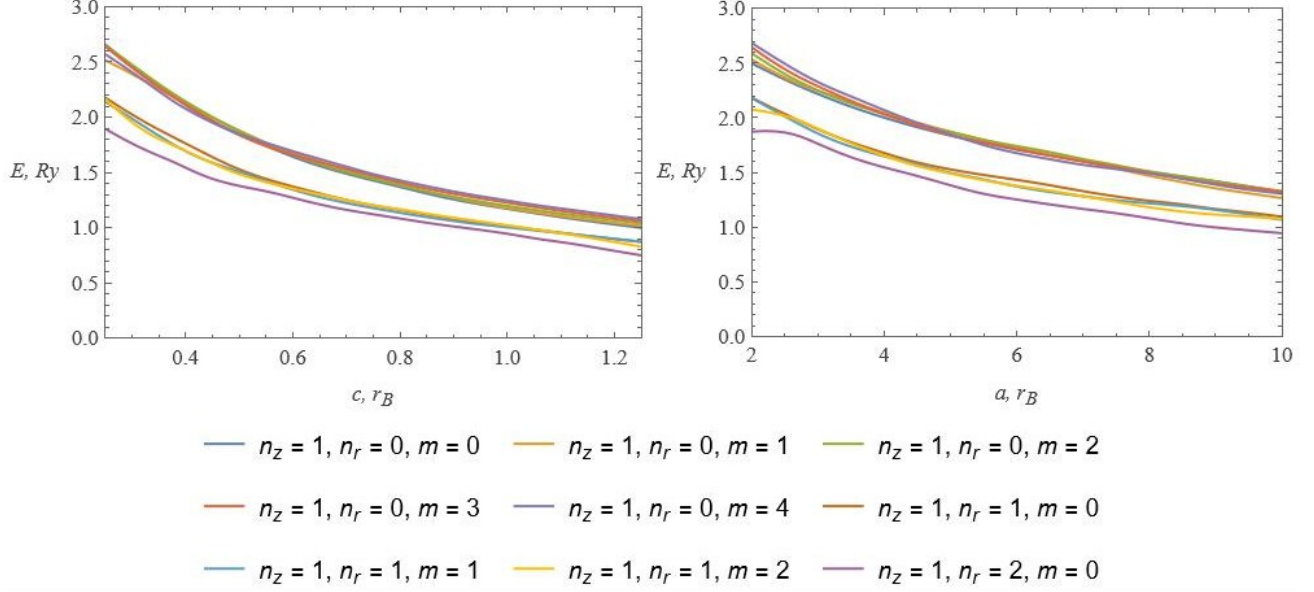


Figure 4. **Exciton Coulomb interaction versus dot size for $n_z = 1$.** **Left:** dependence on the vertical semi-axis c at fixed $a = 5r_B$. **Right:** dependence on the lateral semi-axis a at fixed $c = 0.5r_B$.

The calculated photoluminescence (PL) spectrum $R(\hbar\omega)$ for the representative dot is shown in Figure 5. Each line corresponds to a bright exciton transition with $n_z = 1$ and matched in-plane quantum numbers. The Lorentzian width is set to $\gamma = 0.5$ meV. The sequence of peaks follows the ordering of the single-particle ladder corrected by the electron-hole attraction consistent with the selection rules (20). In the non-interacting limit, the states within a fixed $N \equiv 2n_r + |m|$ would be degenerate with multiplicity $N+1$. The Coulomb term V_{eh} depends on $(n_r, |m|)$ and therefore lifts this degeneracy: for a given N , levels with larger central localization experience stronger binding and shift to lower energies, producing the observed splittings in the PL. Relative peak heights reflect the van Roosbroeck-Shockley factor evaluated at $T = 300$ K.

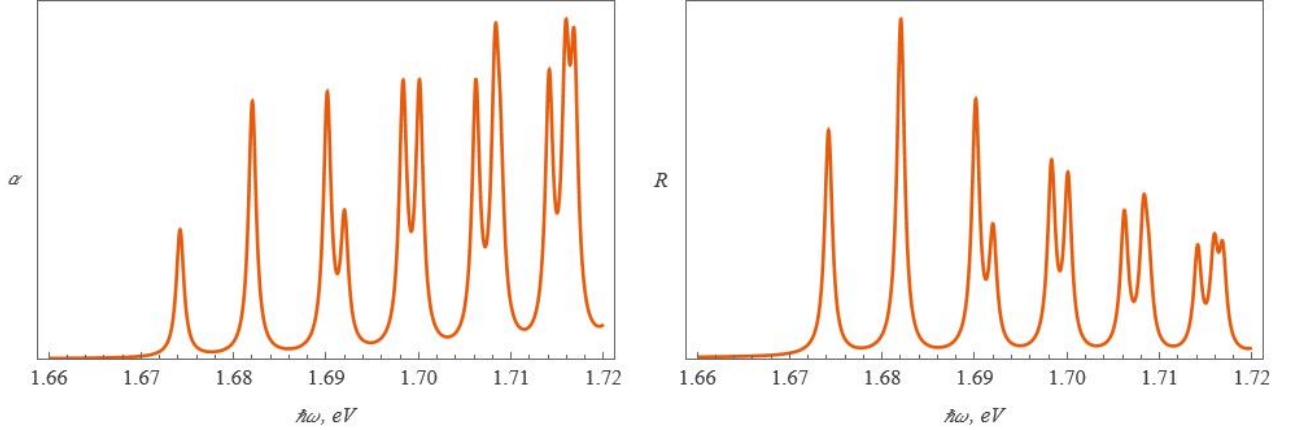


Figure 5. **Absorption and photoluminescence spectra** for a lens-shaped dot with $a = 5r_B$ and $c = 0.5r_B$. **Left:** interband absorption coefficient $\alpha(\hbar\omega)$. **Right:** photoluminescence $R(\hbar\omega)$ computed via the van Roosbroeck-Shockley relation at thermal equilibrium ($T = 300$ K). Both panels use Lorentzian peak broadening with half-width $\gamma = 0.5$ meV.

4. CONCLUSION

We developed an analytic-numerical framework for carriers and excitons in a lens-shaped (semi-ellipsoidal) quantum dot with hard-wall confinement. Within an adiabatic separation of fast (longitudinal) and slow (in-plane)

motion, we derived closed-form envelope functions and the single-particle spectrum, and established simple size and mass dependences. Using these states, we evaluated the electron-hole interaction energy and constructed interband absorption and photoluminescence (PL) spectra.

The main findings are as follows. (i) For the single-particle problem, the energies are much more sensitive to the dot height c than to the lateral semi-axis a ; states with $n_z > 1$ lie well above the $n_z = 1$ manifold and are negligible for optical response in the explored size range. (ii) The Coulomb binding $|V_{eh}|$ decreases monotonically with increasing a or c ; over the plotted ranges, its relative sensitivity to changes in a and c is comparable, and the curves group primarily by the radial quantum number n_r , reflecting the role of central localization. (iii) The absorption and PL spectra are dominated by bright transitions with matched envelope quantum numbers; electron-hole attraction lifts the non-interacting degeneracy within each $N = 2n_r + |m|$ multiplet, shifting more localized states to lower energy and producing the observed multiplet structure. (iv) Radiative lifetimes obtained from the oscillator-strength formalism fall in the picosecond range for the considered parameters and vary smoothly with size.

The approach is deliberately minimal: it assumes infinite barriers, a single isotropic effective mass for each carrier, no strain or piezoelectric fields, no dielectric mismatch at the interface, and neglects valence-band mixing and nonparabolicity. These idealizations make the trends transparent but set limits on quantitative accuracy for specific materials and dots.

Natural extensions include replacing hard walls with finite barriers, adding dielectric mismatch and image-charge effects, incorporating strain and piezoelectric fields for lattice-mismatched systems, and treating heavy-/light-hole mixing. Temperature-dependent broadening and phonon sidebands could be included for a more complete PL lineshape. On the excitonic side, biexcitons and charged complexes, as well as polarization-resolved selection rules, are direct next steps. Finally, fitting the present model to measured spectra for individual dots would enable extraction of geometric parameters and material constants with uncertainty estimates.

In summary, the semi-ellipsoidal model combined with adiabatic separation provides compact analytic states, clear size-dependence trends, and physically interpretable optical spectra, offering a useful baseline for both the design and analysis of lens-shaped quantum dots.

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